Preliminary Data Quality Assessment

Project: Telco Customer Churn Prediction

Dataset: Telco Customer Churn

Project Phase: Data Ingestion and Initial Assessment

Assessment Status: Preliminary — Prior to Data Cleaning

1. Executive Summary

This assessment evaluates the initial quality, structure, and completeness of the Telco Customer Churn dataset before data cleaning and machine learning development.

The dataset contains 7,043 customer records and 21 variables describing customer demographics, subscribed services, account characteristics, billing information, and churn status.

The preliminary assessment identified no duplicate records, duplicate customer identifiers, or standard null values. However, 11 records contain blank strings in the TotalCharges field, preventing the column from being interpreted as numeric upon import.

The dataset also exhibits an imbalanced target distribution, with approximately 26.53% of customers classified as having churned.

Overall, the dataset appears suitable for the intended predictive modeling project, provided the identified data-quality issues are addressed during preprocessing.

2. Assessment Objectives

The preliminary assessment was conducted to:

- Confirm successful ingestion of the source dataset.
- Verify the dataset's dimensions and structure.
- Identify duplicate records and customer identifiers.
- Detect missing values and improperly formatted entries.
- Examine variable data types.
- Establish the initial distribution of the target variable.
- Document findings requiring further investigation or remediation.

This assessment is limited to initial data profiling. No records were modified, removed, or imputed.

3. Dataset Overview

3.1 Dataset Description

The Telco Customer Churn dataset contains historical customer information intended to support the prediction of customer attrition.

The dataset includes demographic characteristics, service subscriptions, contractual arrangements, billing information, and whether each customer discontinued service.

The dependent variable, Churn, indicates customer attrition and will serve as the target for the classification models developed later in the project.

3.2 Dataset Characteristics

Characteristic	Observed Value
Source	Telco Customer Churn dataset, distributed through Kaggle
File format	CSV
Total records	7,043
Total columns	21
Unique customer IDs	7,043
Target variable	Churn
Target categories	Yes, No
Assessment environment	Python / Pandas

The original source file is retained without modification to preserve data integrity and facilitate reproducibility.

4. Structural Integrity Assessment

4.1 Record and Column Validation

The imported dataset contains 7,043 rows and 21 columns, consistent with the dimensions specified in the project implementation guide. The successful import and expected dimensions indicate that the dataset was loaded correctly at the structural level.

4.2 Duplicate Records

Duplicate detection was performed using Pandas.

Validation Check	Result	Assessment
Duplicate rows	0	Passed
Duplicate customer identifiers	0	Passed
Unique customer identifiers	7,043	Passed

No duplicate records or customer identifiers were identified.

The customerID variable uniquely identifies each customer and should be preserved for record tracking and verification. However, it should be excluded from the predictive feature set because it functions as an identifier rather than a meaningful customer characteristic.

5. Completeness and Missing-Value Assessment

5.1 Standard Missing Values

An initial assessment using the Pandas isna() function identified zero standard null values across the dataset.

However, the absence of null values does not establish complete data integrity because improperly formatted strings and placeholder values may not be recognized as missing.

5.2 Blank TotalCharges Values

Further examination identified 11 records containing blank strings in the TotalCharges column.

All 11 affected records correspond to customers with a tenure of zero months.

This relationship suggests that the blank values may be associated with recently established customer accounts that have not yet accumulated billing charges.

Replacing these blanks with zero is a reasonable proposed treatment, provided the assumption is documented.

6. Data Type Assessment

Initial inspection indicates that most variables are represented using data types appropriate to their contents. However, TotalCharges, which represents a monetary amount, is imported as an object rather than a numeric variable.

Variable	Current Data Type	Expected Analytical Type	Assessment
customerID	object	Identifier/string	Appropriate
tenure	int64	Integer	Appropriate
MonthlyCharges	float64	Numeric	Appropriate
TotalCharges	object	Numeric	Requires correction
Churn	object	Categorical/binary	Requires encoding for modeling

The TotalCharges type discrepancy results from the presence of blank strings within an otherwise numeric column.

Other categorical variables will require appropriate encoding before machine learning model training. These transformations are part of the later preprocessing stage and are not classified as source/data errors.

7. Target Variable Assessment

7.1 Churn Distribution

The target variable, Churn, contains two categories:

Churn Status	Customer Count	Percentage
No — Retained	5,174	73.47%
Yes — Churned	1,869	26.53%
Total	7,043	100.00%

Approximately one-quarter of the customers in the dataset are classified as having churned.

7.2 Modeling Implications

The distribution indicates a moderate class imbalance, with substantially more retained customers than churned customers.

This imbalance does not represent a data/quality defect. However, it introduces considerations for predictive model development and performance evaluation.

A model that consistently predicts customer retention could achieve approximately 73.47% accuracy without successfully identifying any customers who churn.

Consequently, evaluation should extend beyond accuracy to include metrics such as precision, recall, F1-score, and the confusion matrix. Stratified sampling should also be considered when separating the dataset into training and testing subsets.

8. Data Quality Findings and Risk Assessment

The following table summarizes the primary findings identified during the preliminary assessment.

Finding	Severity	Potential Impact	Recommended Action
Blank TotalCharges entries	Medium	Missing billing information for 11 customers	Investigate and impute where justified
TotalCharges stored as object	Medium	Prevents direct numerical analysis	Convert to numeric
customerID included in dataset	Low	May introduce an irrelevant modeling feature	Retain for tracking; exclude from training
Imbalanced churn distribution	Modeling consideration	Accuracy alone may provide misleading evaluation	Use stratification and appropriate metrics
Duplicate records	None identified	No issue detected	No remediation required